SOMUN 2026

BACKGROUND GUIDE

United Nations Office for Outer Space Affairs (UNOOSA)



Agenda: Enhancing Orbital Safety and Space Traffic Management: Mitigating Space Debris Risks and Regulating Dual-Use Rendezvous and Proximity Operations under the Long-Term Sustainability Framework

1. Letter from the executive board:

Dear Delegates,

It is with great honour that we welcome you to the United Nations Office for Outer Space Affairs. We are honoured to serve as your Executive Board and look forward to three days of thoughtful deliberation, rigorous negotiation, and meaningful diplomacy.

Unlike a lot of MUN committees, UNOOSA isn't about dramatic speeches or talking over people to win an argument. It really comes down to understanding the technical realities, getting the legal details right, and putting in the hard work to reach consensus. Space isn't some far-off frontier anymore, it quietly runs our daily lives down here. The problems we're looking at, like crowded orbits, dangerous space junk, and the blurry line between cleaning up satellites and weaponizing them, need actual, workable answers instead of vague promises.

We put this background guide together to give you the basic context you need to kick off your preparation. Think of it as a launching pad rather than a complete textbook. We expect you to dig deeper on your own, look up the actual treaties, read your country's domestic space policies, and build an argument you can genuinely back up. Just keep in mind that you can't cite this guide itself as official evidence during debate.

You don't have to show up on day one with everything figured out. What we care about most are delegates who think critically, back up their points with facts, and are ready to actually listen to the other side. Good diplomacy in our room isn't about who gives the most speeches or speaks the loudest, it's about who can take completely opposite national interests and find a way to make them work together.

We remain available to assist you throughout your journey leading up to the conference, so please feel free to reach out. We wish you the very best in your preparations and look forward to an engaging committee conference.

Warm regards,

Chairperson	***Vice Chairperson***	***Rapporteur***
Sarvagna T	*Risheeth B*	*Arul Ch*

2. Important points to remember:

1. Procedure: The purpose of putting in procedural rules in any committee is to ensure a more organized and efficient debate. The committee will follow the UNA-USA Rules of Procedure. Although the Executive Board shall be fairly strict with the Rules of Procedure, the discussion of the agenda will be the main priority. So, delegates are advised not to restrict their statements due to hesitation regarding procedure.

2. Foreign Policy: Following the foreign policy of one's country is the most important aspect of a Model UN Conference. This is what essentially differentiates a Model UN from other debating formats. To violate one's foreign policy without adequate reason is one of the worst mistakes a delegate can make.

3. Role of the Executive Board: The Executive Board is appointed to facilitate debate. The committee shall decide the direction and flow of debate. The delegates are the ones who constitute the committee and hence must be uninhibited while presenting their opinions/stance on any issue. However, the Executive Board may put forward questions and/or ask for clarifications at all points of time to further debate and test participants.

4. Nature of Source/Evidence: This Background Guide is meant solely for research purposes and must not be cited as evidence to substantiate statements made during the conference. Evidence or proof for substantiating statements made during formal debate is acceptable from the following sources:

a. **United Nations:** Documents and findings by the United Nations or any related UN body is held as a credible proof to support a claim or argument. Multilateral Organizations: Documents from international organizations like OIC, NATO, SAARC, BRICS, EU, ASEAN, the International Court of Justice, etc. may also be presented as credible sources of information.
b. **Government Reports:** These reports can be used in a similar way as the State Operated News Agencies reports and can, in all circumstances, be denied by another country.
c. **News Sources:**
    1. Reuters: Any Reuters article that clearly makes mention of the fact or is in contradiction of the fact being stated by a delegate in council.
    2. State operated News Agencies: These reports can be used in the support of or against the State that owns the News Agency. These reports, if credible or substantial enough, can be used in support of or against any country as such but in that situation, may be denied by any other country in the council. Some examples are RIA Novosti (Russian Federation), Xinhua News Agency (People's Republic of China), etc.

Please Note: Reports from NGOs working with UNESCO, UNICEF and other UN bodies will be accepted. Under no circumstances will sources like Wikipedia be accepted.

3. Hierarchy of evidence:

Evidence can be presented from a wide variety of sources, but not all sources are treated as equal. Here's the hierarchy in which evidence is categorised:

Tier 1: Includes any publication, statement, resolution, or document released by any of the Nations' official organs or committees; any publication, statement, or document released by a UN member state in its own capacity. The evidence falling in this tier is considered most reliable during the simulation.

Tier 2: Includes: any news article published by any official media source that is owned and controlled by a UN member state. E.g.: Xinhua News (China), Prasar Bharti (India), BBC (United Kingdom) etc. The evidence falling in this tier is considered sufficiently reliable in case no other evidence from any Tier 1 source is available on that particular fact, event, or situation.

Tier 3: Includes: any publication from news sources of international repute such as Reuters, The New York Times, Agence-France Presse, etc. The evidence falling under this tier is considered the least reliable for the purposes of this simulation. Yet, if no better source is available in a certain scenario, it may be considered.

4. Introduction to the committee:

Based in Vienna, the United Nations Office for Outer Space Affairs (UNOOSA) is essentially the UN's point team for everything orbiting above us. Its main mission is keeping space peaceful, usable, and fair for everyone, especially newer spacefaring nations that are trying to set up their own space programs and national laws without getting dominated by the legacy powers.

Behind the scenes, UNOOSA handles the actual day-to-day work. It acts as the secretariat for the Committee on the Peaceful Uses of Outer Space (COPUOS) and its Scientific, Technical, and Legal Subcommittees. It also maintains the official UN Register of Objects Launched into Outer Space, logging what's going up under the 1975 Registration Convention, and runs practical networks like UN-SPIDER, which routes satellite data to emergency teams dealing with natural disasters on the ground. Put together, that makes UNOOSA the central hub where international space law, orbital traffic safety, and real space diplomacy actually come together.

Before you get too deep into preparation, there's an important detail about how our room is structured.

In the real world, the United Nations Office for Outer Space Affairs (UNOOSA) isn't a debating body. It's an administrative office staffed by UN civil servants,

which means countries don't actually sit there to give speeches, vote, or pass resolutions. Because of that, UNOOSA cannot be simulated in an MUN like a regular committee.

Instead, over the next few days, we'll be operating as the Committee on the Peaceful Uses of Outer Space (COPUOS). COPUOS is the actual intergovernmental body where national diplomats sit down, deliberate international space law, and negotiate agreements, with UNOOSA serving as the permanent secretariat in the background supporting the work.

So, when you step into the room, you'll be representing your country as an official delegate to COPUOS: driving policy, forming blocs, and working toward consensus.

5. COPUOS:

The Committee on the Peaceful Uses of Outer Space (COPUOS) was set up by the General Assembly in 1959 to govern the exploration and use of space for the benefit of all humanity: for peace, security and development. The Committee was tasked with reviewing international cooperation in peaceful uses of outer space, studying space-related activities that could be undertaken by the United Nations, encouraging space research programmes, and studying legal problems arising from the exploration of outer space.

The Committee was instrumental in the creation of the five treaties and five principles of outer space. International cooperation in space exploration and the use of space technology applications to meet global development goals are discussed in the Committee every year. Owing to rapid advances in space technology, the space agenda is constantly evolving. The Committee therefore provides a unique platform at the global level to monitor and discuss these developments.

The Committee has two subsidiary bodies: the Scientific and Technical Subcommittee, and the Legal Subcommittee, both established in 1961. The Committee reports to the Fourth Committee of the General Assembly, which adopts an annual resolution on international cooperation in the peaceful uses of outer space.

Since its establishment, the Committee's membership has continued to expand. The Committee is the only committee of the General Assembly dealing exclusively with international cooperation in the peaceful uses of outer space, and its role as a forum to monitor and discuss developments related to the exploration and use of outer space has evolved alongside with the technical advancements in space exploration, geopolitical changes, and the evolving use of space science and technology for sustainable development.

The Scientific and Technical Sub-committee: The Scientific and Technical Subcommittee (STSC) meets every year for two weeks to discuss questions related to the scientific and

technical aspects of space activities. Topics for discussion include space weather, near-Earth objects, the use of space technology for socioeconomic development, or for disaster management support, global navigation satellite systems, and the long-term sustainability of outer space activities.

The Legal Sub-committee: The Legal Subcommittee meets every year for two weeks to discuss legal questions related to the exploration and use of outer space. Topics include the status and application of the five United Nations treaties on outer space, the definition and delimitation of outer space, national space legislation, legal mechanisms relating to space debris mitigation, and international mechanisms for cooperation in the peaceful exploration and use of outer space.

The COPUOS Plenary is the main room where everything in international space policy actually comes together. Earlier in the year, delegates break off into two specialized bodies: the Scientific and Technical Subcommittee handles the math, physics, and engineering challenges, while the Legal Subcommittee deals with treaties and international law. The Plenary is where both sides sit at the same table to turn technical data and legal principles into a unified policy before sending it to the UN General Assembly. The room runs on strict consensus. You can't just cobble together a 2/3rd majority to push a resolution through; if even one delegation has an objection, the text stalls. Success in this committee isn't about winning a vote, it's about finding practical solutions that every country in the room can actually sign on.
The committee will be simulating COPUOS-Plenary.

6. How this committee works?

How This Committee Actually Works:
To keep things super clear from day one: even though your schedule says UNOOSA, you won’t actually be playing the role of UNOOSA staff. In real life, UNOOSA is basically the UN's backstage administrative team; they handle satellite paperwork, keep the orbital registries up to date, and route emergency data down to Earth. They don't give speeches, push national agendas, or vote. Instead, you will be sitting as official government diplomats in COPUOS (the Committee on the Peaceful Uses of Outer Space), specifically in its Plenary session. In the real UN, COPUOS breaks off into two study groups earlier in the year: the Scientific & Technical team (who obsess over rocket math, debris tracking, and physics) and the Legal team (who dissect treaties and liability). The Plenary is where both sides finally sit in the same room to turn the science and the law into one coherent deal. The biggest catch? This room runs on strict consensus. You can’t just rally a quick majority to drown out people you disagree with. If even one country has a major problem with your draft, the whole thing stalls. That means your goal

isn't to beat the other side in a vote, it's to talk it out until you build a deal everyone can actually live with.

A Quick Breakdown (FAQs about the committee)

1.Who am I supposed to be in here?
You are an official diplomat speaking directly for your assigned country. You fight for your government's space program, your economy, and your national security.

2.What does "Plenary" even mean, and why are we doing this?
Think of the subcommittees as two separate breakout rooms: one that only talks about engineering and another that only talks about legal clauses. The Plenary is just the big meeting where everyone comes back together to make actual decisions. We chose the Plenary so you don't get boxed in, you can argue about orbital physics and international treaties in the very same speech without anyone telling you it's out of order.

3.Can my bloc just team up and outvote the room?
Nope. COPUOS doesn't do special majority rule. Every delegate in this room effectively holds a veto. If your bloc drafts a resolution that sidelines another major space player or ignores smaller developing nations, they can simply say "I object," and the draft goes nowhere.

4.How do we pass a resolution?
Old-school diplomacy. You listen to what other countries genuinely care about, and find practical middle ground and pass a resolution with the consensus of the entire committee.

8. Introduction of agenda:

For decades, countries launched satellites with the mindset that outer space was practically infinite, countries put their payload up, ran their mission, and left the hardware behind when it died. That approach simply does not work anymore. Low Earth Orbit has become essential day-to-day infrastructure, quietly keeping maps working, weather alerts running, flights navigating safely, and financial markets connected.

The issue is that orbit is getting crowded fast. Tens of thousands of operational satellites are now sharing narrow orbital lanes with old rocket boosters and millions of untracked fragments of debris. At orbital speeds over 27,000 kilometers per hour, even a stray screw hits like high-grade artillery. A major crash does not just claim one satellite; it scatters thousands of new high-speed fragments, raising the risk of cascading collisions, often referred to as the Kessler syndrome.

To keep these pathways open, the world has focused on two practical fixes: setting up proper Space Traffic Management and building Active Debris Removal tech to grab dead hulls and pull them down.

Turning those fixes into actual international policy, however, runs straight into tough political and legal friction:

- **No Real Traffic Controller:** Planes don't run into each other because international air traffic controllers talk around the clock. Space has nothing like that. Right now, operators dodge near-misses using a mix of incomplete military catalogs, private tracking data, and ad-hoc phone calls across time zones.
- **The Trust Problem with RPO:** To haul away dead debris or refuel a satellite, you need Rendezvous and Proximity Operations (RPO)-the ability to steer close and latch onto another object. The catch is that the exact same engineering can be used to spy on, blind, or smash an adversary's satellite. Because it's impossible to tell what a maneuvering satellite is thinking, an unannounced close approach easily looks like a military threat.
- **Legal Red Tape on Space Trash:** Under the 1967 Outer Space Treaty, a country owns its spacecraft forever. That means even if an abandoned, out-of-control rocket body is on track to smash into something, you cannot touch or remove it without explicit permission from the nation that launched it.
- **Fairness for Newer Space Programs:** Legacy space powers filled the best orbits first and now want tighter, more expensive de-orbit rules. Emerging space nations have valid concerns that these steep compliance costs will make it harder financially to launch their programs.

Working under the UN's Long-Term Sustainability (LTS) Guidelines, our job in this room isn't to trade idealistic speeches. It's to find the practical compromises that keep orbit physically safe without shutting developing nations out or compromising anyone's security.

Key Terms Breakdown

- **Orbital Safety:** Keeping Earth's orbit physically usable and collision-free. Because objects circle the planet at over 27,000 km/h, avoiding crashes, fuel tank explosions, and uncontrolled re-entries is vital to protecting the daily satellite services society relies on.
- **Space Traffic Management (STM):** The rules and coordination systems needed to stop spacecraft from running into each other. Unlike civil aviation, space has no central air traffic controller; today, operators mostly rely on a patchwork of military radar alerts, private tracking data, and direct emails to dodge collisions.
- **Space Debris Risks:** The hazard posed by dead hardware, abandoned rocket stages, and millions of untracked fragments circling Earth. At orbital speeds, even a loose screw hits like an exploding grenade, and major crashes risk triggering the **Kessler syndrome**—a chain reaction of collisions that could choke off vital orbits for generations.

- **Dual-Use:** Technology designed for peaceful, civilian purposes that can double as a space weapon. A robotic arm built to refuel a satellite or grab space junk can just as easily rip off an adversary's antenna, making it almost impossible to tell helpful maintenance apart from a covert attack.
- **Rendezvous and Proximity Operations (RPO):** The maneuver of deliberately steering one spacecraft close to another to inspect, dock, or refuel it. While essential for clearing debris and repairing payloads, unannounced close passes near foreign satellites look like military threats, sparking severe geopolitical suspicion.
- **Long-Term Sustainability (LTS) Framework:** A set of 21 practical guidelines adopted by COPUOS in 2019 to keep space safe, clean, and accessible for future generations.

9. Historical evolution of the agenda:

1. The Cold War Rulebook (1960s–1970s): Writing the Basics

When Sputnik launched in 1957, countries panicked that orbit would quickly turn into a nuclear playground. The United Nations stepped in to build a legal foundation before things got out of hand:

- **The 1967 Outer Space Treaty (OST):** The core constitution of space. It declared outer space the "province of all mankind," banned nukes in orbit, made land claims illegal, and held governments responsible for anything their citizens or companies launch. Crucially, Article VIII gave countries permanent ownership and control over their space objects.
- **The Follow-Up Treaties (1972 & 1975):** The Liability Convention created rules for who pays if a falling spacecraft damages property on Earth, while the Registration Convention set up a basic registry so the UN knows what goes up.

The Catch: These treaties were built on two outdated assumptions: that only a handful of wealthy superpowers would ever launch rockets, and that space was an infinite vacuum where nobody would ever run out of room. They contained zero rules about clearing out junk or managing satellite traffic.

2. The Kessler Warning (Late 1970s–1990s): Recognizing the Trash Problem

In 1978, NASA scientist Donald Kessler ran the numbers and came away with a terrifying conclusion: if orbit gets too crowded, collisions will start creating thousands of fragments that crash into other satellites, kicking off an unstoppable domino effect of crashes. This nightmare scenario became known as the **Kessler syndrome**.

At first, people brushed it off as science fiction. But by the late 1980s, abandoned upper rocket stages were literally blowing up in orbit on their own because leftover fuel expanded and ruptured their tanks. In 1993, space agencies formed the **Inter-Agency Space Debris Coordination Committee (IADC)** to find practical engineering fixes, like requiring rockets to

vent their unspent fuel and discharge batteries at the end of their mission so they wouldn't turn into orbital bombs.

3. The Wake-Up Calls (2000s): Theory Becomes Reality

In the 2000s, debris stopped being a theoretical concern and became an active crisis:

- **The 2007 Chinese Anti-Satellite Test:** China fired a missile to blow up its own aging weather satellite (*Fengyun-1C*). The blast created over 3,500 trackable shards of shrapnel, the single messiest debris event in human history, polluting orbits that satellites still use today.
- **The 2009 Iridium-Cosmos Crash:** Over Siberia, an active commercial phone satellite (*Iridium 33*) slammed directly into a dead Russian military satellite(*Kosmos-2251*) at roughly 42,000 km/h. It blew both spacecraft into thousands of high-speed bullets and proved once and for all that satellites don't just dodge each other on luck. Without coordinated traffic rules, major orbital crashes were inevitable.

4. The Mega-Constellation Boom and the LTS Guidelines (2010s)

The 2010s completely flipped space economics. Private launch companies brought costs down, leading to the rise of "mega-constellations" fleets of thousands of mass-produced commercial satellites launched into the exact same low Earth orbital shells.

Realizing that Cold War-era treaties couldn't handle thousands of commercial satellites, COPUOS spent nearly a decade laying out the **Guidelines for the Long-Term Sustainability of Outer Space Activities (LTS)**, formally adopted in 2019. These 21 guidelines gave the world practical recommendations on sharing tracking data, registering payloads properly, and keeping space clean. The catch? They are voluntary "soft law," meaning they only work if individual governments choose to enforce them in their national legislatures.

5. Where We Are Now: Cleanup Tech Meets Military Paranoia

Today, relying on dead satellites to slowly fall back into the atmosphere and burn up on their own isn't fast enough to keep orbits clear. The focus has rapidly pivoted to **Active Debris Removal,** sending up space operations with robotic arms and nets to physically grab dead rocket hulls and drag them down.

That brings us directly to the core friction of this agenda:

- **The Trust Trap (Dual-Use RPO):** The exact maneuvers required to steer up to a dead rocket and grab it (**Rendezvous and Proximity Operations**) are identical to the maneuvers an attack satellite would use to sneak up, hack, blind, or physically smash an adversary's military satellite. Because intent is invisible from the ground, unannounced close maneuvers trigger immediate military panic.

- **The Legal Stalemate:** Because Article VIII of the Outer Space Treaty gives a nation permanent ownership over its spacecraft, you legally cannot sweep up another country's abandoned, drifting rocket booster without their permission—even if that dead booster is an immediate hazard to everyone else in orbit.

10. Existing international frameworks:

1. The Hard Law: The UN Space Treaties (Legally Binding):

These treaties were negotiated through COPUOS and adopted by the UN General Assembly. Under international treaty law, states that have signed and ratified them are legally required to comply with their terms.

The Outer Space Treaty (OST, 1967)

Often regarded as the foundational constitution of outer space, the OST sets down the ground rules for all human activities beyond Earth's atmosphere:

- **Article I (The Common Interest of Humanity):** Establishes that the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development, and shall remain the "province of all mankind".
- **Article II (Non-Appropriation):** Prohibits national appropriation by claim of sovereignty, by means of use or occupation, or by any other means. No country can claim an orbit, the Moon, or any celestial body as its sovereign territory.
- **Article VI (State Responsibility & Supervision):** Holds governments directly responsible under international law for all national space activities, whether conducted by governmental agencies or non-governmental commercial entities. States must formally authorize and continuously supervise their private space operators.
- **Article VIII (Jurisdiction and Ownership):** Establishes that the state that registers a space object retains perpetual jurisdiction, control, and ownership over that object and any component parts wherever it travels in outer space or returns to Earth.
- **Article IX (Cooperation and "Due Regard"):** Obligates states to conduct all space activities with due regard to the corresponding interests of all other states, and requires international consultations prior to conducting any activity that might cause potentially harmful interference.

The Liability Convention (1972)

This convention creates an international law for assigning financial responsibility when a space object causes physical damage or injury:

- **Absolute Liability for Surface and Air Damage:** If a spacecraft, rocket body, or re-entering fragment causes damage on the surface of the Earth or to aircraft in flight, the

launching state is held absolutely liable to pay financial compensation without requiring proof of negligence.
- **Fault-Based Liability in Orbit:** If an incident occurs between two spacecraft in outer space, liability is determined based on the fault of the state or persons for whom the state is responsible.

The Registration Convention (1975)

To ensure accountability and transparency, this treaty establishes a standardized system for identifying spacecraft:

- **The UN Register:** Requires every launching state to maintain a national registry and transmit specific orbital parameters, such as the date of launch, territory or facility of launch, basic orbital characteristics (nodal period, inclination, apogee, and perigee), and general function to the Secretary-General of the United Nations (managed by UNOOSA).

2. The Soft Law: Technical Guidelines (Voluntary):

Because modern space operations move faster than traditional treaty negotiations, states rely heavily on non-binding, consensus-driven technical guidelines to coordinate day-to-day orbital practices.

The IADC Space Debris Mitigation Guidelines (2002 / UN Endorsed in 2007)

Originally formulated by the Inter-Agency Space Debris Coordination Committee (IADC) and later adopted by COPUOS and the UN General Assembly, this document outlines operational engineering practices to curb debris generation:

- **Mission Debris Minimization:** Encouraging manufacturers to design rockets and satellites that do not intentionally release separation bolts, lens covers, or strap-on hardware into long-term orbit.
- **Passivation:** Mandating that operators deplete or vent all residual onboard energy sources, including unspent propellant tanks and pressurized battery cells, at the conclusion of a mission to prevent spontaneous in-orbit explosions.
- **Post-Mission Disposal:** Recommending that spacecraft in Low Earth Orbit (LEO) maneuver into disposal orbits that naturally re-enter Earth's atmosphere within 25 years, while satellites in Geostationary Orbit (GEO) maneuver into a designated "graveyard orbit" safely above operational lanes.

The COPUOS Long-Term Sustainability (LTS) Guidelines (2019)

The central reference document for our agenda, the LTS Framework consists of **21 agreed guidelines** designed to ensure the safe, peaceful, and sustainable use of outer space over the coming decades. They are organized across four pillars:

- **Pillar A: Policy and Regulatory Frameworks:** Guides governments on updating their national space laws, establishing domestic licensing systems, supervising commercial actors, and ensuring responsible frequency management.

- **Pillar B: Safety of Space Operations:** Outlines operational measures such as sharing Space Situational Awareness (SSA) tracking data, improving conjunction assessment (collision warning) procedures, and performing pre-launch safety coordination.
- **Pillar C: International Cooperation, Capacity-Building, and Awareness:** Encourages established space nations to share technical expertise, tools, and best practices with emerging and developing space programs so orbital safety standards remain universally accessible.
- **Pillar D: Scientific and Technical Research:** Supports ongoing research into orbital modeling, new debris monitoring techniques, space weather monitoring, and the development of next-generation technologies for safe de-orbiting and active remediation.

The Working Group on the Long-Term Sustainability of Outer Space Activities (WG-LTS):

The Working Group on the Long-Term Sustainability of Outer Space Activities (WG-LTS) is essentially the UN's main engine for keeping space usable and safe for the long haul. After spending nearly ten years hammering out the 21 voluntary LTS Guidelines adopted in 2019, the team didn't just leave. Under its current phase, often called "LTS 2.0", the group tracks whether countries are actually writing those safety guidelines into their own national laws. It also tackles new problems that weren't as massive a decade ago, like thousands of commercial satellites crowding low Earth orbit and blocking our view of the stars. At the same time, the group acts as a bridge for emerging space nations: it works to share tracking data and technical support so developing programs can launch satellites safely without getting priced out by strict, expensive rules.

11. Challenges to the agenda:

1. The Debris Chain Reaction and the Kessler Nightmare

- **The Reality in Orbit:**
  Low Earth Orbit is packed with old rocket boosters, dead satellites, and millions of untracked shards of metal whizzing around at over 27,000 km/h. At those speeds, even a loose screw hits like an exploding grenade, and basic shielding won't save you. The nightmare scenario is the **Kessler syndrome**: an orbital domino effect where one crash blasts apart into thousands of high-speed bullets that shatter other satellites, turning critical orbital pathways into unusable junkyards for generations.
- **Why the Law Fails:**
  - **The "25-Year Rule" Is Stuck in the Past:** Decades ago, guidelines suggested letting dead satellites take up to 25 years to fall and burn up naturally in the atmosphere. That made sense when launches were rare events. But with thousands of mass-produced satellites going up every year, waiting a quarter of a century creates a dangerous orbital traffic jam.
  - **No Real Teeth:** The UN's 2019 Long-Term Sustainability (LTS) Guidelines give solid, practical advice, but they are purely voluntary. Unless national governments

turn them into binding conditions for commercial launch licenses, companies can skip costly de-orbit systems to save money without facing legal consequences.

2. The Cleanup Trap: Active Removal vs. Sovereign Ownership

- **The Reality in Orbit:**
  Waiting for space junk to fall down on its own isn't fast enough anymore. Scientists agree we need space tugs to fly up, latch onto dead, multi-ton rocket boosters, and drag them down to burn up. But the moment you try to touch another nation's hardware, you run straight into a geopolitical wall.
- **Why the Law Fails:**
  - **You Can't Touch Other People's Junk (OST Article VIII):** Under Article VIII of the Outer Space Treaty, a country owns and controls its space objects forever. Hardware never becomes "abandoned property." Even if a derelict booster is tumbling out of control toward an active satellite fleet, no other country or company can legally touch it without explicit permission.
  - **No Salvage Rights in Space:** At sea, maritime law allows sailors to salvage or clear abandoned, hazardous shipwrecks in international waters. Space law has no equivalent. If the original launching state ignores calls or no longer exists, that lethal hazard stays legally untouchable.
  - **The Penalty for Helping:** If a private cleanup tug tries to capture a dead foreign rocket and the maneuver goes wrong, shattering the booster into thousands of new fragments, the 1972 Liability Convention holds the cleaning nation liable. Taking the initiative to clean up orbit comes with massive financial and legal risks.

3. Space Traffic Management: Flying Blind with No Right-of-Way

- **The Reality in Orbit:**
  Aviation works because air traffic controllers monitor flight paths around the clock and tell pilots who need to climb, turn, or wait. Space has nothing like that. Right now, satellite operators dodge near-misses using a fragmented mix of military radar catalogs, private tracking feeds, and frantic direct emails sent across time zones.
- **Why the Law Fails:**
  - **No proper rules:** Under the 1972 Liability Convention, if two satellites crash in orbit, the party at fault has to pay. But space has no speed limits, no flight lanes, and no right-of-way rules dictating who was supposed to burn fuel to get out of the way. Without rules of the road, proving legal negligence in court is difficult.
  - **Outdated Paperwork:** The 1975 Registration Convention only asks countries to log launches "as soon as practicable," which often means months after the fact. Worse, it only notes where a satellite was on day one, not where it maneuvers, changes orbits, or dies, leaving the UN Register useless as a live traffic map.
  - **Secret Radar Feeds:** The LTS Guidelines urge nations to share orbital tracking data. But superpowers treat pinpoint radar and sensor data as top-secret military assets and refuse to share live feeds openly.

4. The Dual-Use Dilemma: Maintenance or Covert Attack?

- **The Reality in Orbit:**
  To fix an active satellite, top off its fuel, or haul down debris, a spacecraft has to perform **Rendezvous and Proximity Operations (RPO)** deliberately flying within meters of another object. The catch is that the technology is inherently dual-use: the robotic arm designed to latch onto space junk can just as easily rip off an adversary's spy satellite antenna.
- **Why the Law Fails:**
  - **No Safe Buffers:** The Outer Space Treaty guarantees free access to space and bans claiming sovereign territory. Because no nation can carve out an exclusive boundary in orbit, international law has no agreed definition for how close a spacecraft can get before it crosses the line from peaceful flight into an active threat.
  - **Invisible Motives:** You can't read a satellite's real intent through a telescope on Earth. Because treaties don't mandate advance flight plans, camera inspections, or broadcast transponders for close-range maneuvers, every unannounced approach looks like an incoming attack.

5. The Fairness Divide: Established Powers vs. Emerging Programs

- **The Reality in Orbit:**
  Legacy space powers and mega-constellation operators have already claimed the most valuable orbital slots. Now that orbit is crowded, those same established nations are pushing for strict, expensive environmental rules.
- **Why the Law Fails:**
  - **Pulling Up the Ladder (OST Article I):** Article I promises that space must be open to every country equally, regardless of economic development. Developing nations worry that pricey new mandates, like compulsory collision-avoidance thrusters, advanced debris shields, and expensive tracking systems will act as barriers, pricing emerging space programs out of orbit to clean up a mess left by legacy players.
  - **Empty Promises of Support:** While the LTS Guidelines talk about technical assistance and capacity-building for developing programs, they offer no dedicated UN funding pool, no mandatory tech transfers, and no financial support to help new space agencies meet those costly standards.

12. Questions before the committee:

- How can countries legally clean up old, dead rocket boosters without getting in trouble for touching another nation’s property?
- If a cleanup mission accidentally breaks a dead satellite into smaller pieces, how do we protect that team from having to pay millions in damages?
- Should the UN force satellite companies to burn up their dead satellites within 5 years instead of 25 years?
- If two satellites are about to crash into each other, who has to burn fuel and move out of the way?

- How can the UN registration list be updated in real time so we know where satellites are moving, not just when they launch?
- How can we get military superpowers to share their secret radar tracking data with a public UN database?
- How close can one satellite get to another before it is seen as an attack, and how do we set safe distances without claiming territory in space?
- What proof (like open camera feeds, flight notices, or radio tags) should be required to show that a close-up mission is just for repairs or cleanup, not spying?
- Space is already crowded by rich nations, how do we pass strict safety rules without making space exploration too expensive for poorer countries?
- How can the UN help emerging space programs get free access to satellite tracking data and safety tools?

13. Relevant links:

UNOOSA

COPUOS

UN Treaties and Principles on Outer Space (Full Text Collection)

UNOOSA Registration Information & Submissions

COPUOS Guidelines for the Long-Term Sustainability of Outer Space Activities (LTS Guidelines)

UNOOSA LTS Resource & Implementation Portal

IADC Space Debris Mitigation Guidelines

Consortium for Execution of Rendezvous and Proximity Operations (CONFERS)

Working Group on the Long-Term Sustainability of Outer Space Activities

Convention on International Liability for Damage Caused by Space Objects (Liability Convention, 1972)

Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space (UNGA Resolution 62/217)

Report of the Committee on the Peaceful Uses of Outer Space (Latest Session)